LESSON 2.7

Divide Decimals Using an Algorithm

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can divide with decimals by making the divisor a whole number first.

WHY THIS MATTERS A school bought new math workbooks as shown in the order form. What is the cost of each workbook?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Dividend *Spanish: Dividendo*	The total number being divided into equal groups (the number inside the division bracket).	In 84.6 ÷ 6, it is 84.6
Divisor *Spanish: Divisor*	The number of equal groups you are dividing into (the number outside the division bracket).	In 84.6 ÷ 6, it is 6
Quotient *Spanish: Cociente*	The answer when you divide.	84.6 ÷ 6 = 14.1
Decimal Division Algorithm *Spanish: Algoritmo de división decimal*	The steps for dividing decimals: slide the divisor's decimal point right until the divisor is a whole number, slide the dividend's point right the same number of places, then divide.	8.4 ÷ 2.1 → move the point in both → 84 ÷ 21 → divide, multiply, subtract, bring down → 4
Equivalent division *Spanish: División equivalente*	Multiplying both numbers by 10 or 100 gives the same answer.	7.56 ÷ 0.12 = 756 ÷ 12 = 63 (both multiplied by 100)
Annex a zero *Spanish: Agregar un cero*	Write a zero after the last decimal digit without changing the value.	7.2 → 7.20

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A rope 25.2 meters long is cut into 6 equal pieces. How long is each piece?

1. Write 25.2 ÷ 6. Put the quotient's point above the point in 25.2.
2. 6 goes into 25 four times, 1 left over. Bring down the 2 to make 12.
3. 12 ÷ 6 = 2. Check: 4.2 × 6 = 25.2.

Answer: 4.2 meters for each piece

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's do 8.4 ÷ 2.1. STEP 0 — MOVE THE POINT: how many places make 2.1 a whole number? (1 place, so 2.1 becomes 21.)

1. _____
2. _____
3. _____
4. _____
5. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The dividend is _____ and the divisor is _____ .
- I placed the decimal point _____ because _____ .
- When a remainder appeared I _____ .
- I checked my quotient by _____ .

Word bank dividend | divisor | quotient | decimal point | algorithm | annex | bring down | place value | estimate | check | multiply | subtract

WATCH OUT!

A common mistake

A common mistake in Divide Decimals is moving the decimal point in the divisor but forgetting to move it the same number of places in the dividend. For example, in 6.4 ÷ 0.8 a student changes 0.8 to 8 but leaves the dividend as 6.4, then divides 6.4 ÷ 8 = 0.8 instead of moving both points to get 64 ÷ 8 = 8. A second version of this mistake happens when the divisor has two decimal places, like 0.12: students move both points only 1 place instead of 2, turning 9.6 ÷ 0.12 into 96 ÷ 1.2 instead of the correct 960 ÷ 12. Count the divisor's decimal places first, then move BOTH points that same number of places.

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Divide Decimals Using an Algorithm

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Rewrite each problem by making the divisor a whole number, then solve.

Original	Multiply Both By	New Problem	Quotient
4.5 ÷ 0.9	10		
3.6 ÷ 0.4	10		

STEPS TO TRY

1. The 'Multiply Both By' column is the key move: choose 10 or 100 based on how many decimal places the DIVISOR has.
2. One decimal place in the divisor means multiply both numbers by 10; then rewrite and divide whole numbers.
3. Row 1: 0.9 has one decimal place, so multiply 4.5 and 0.9 each by 10 and write the new whole-number problem in the row.

SHOW YOUR WORK

PART 2 Practice the skill

show every step

2 **Circle the letter of the best answer. Show how you know.**

What is 14.4 ÷ 1.2?

(A) 0.12
(B) 1.2
(C) 12
(D) 120

STEPS TO TRY

1. Dividing by a decimal is awkward — notice the divisor 1.2 has one decimal place. There's a move that makes it whole.
2. Multiply BOTH numbers by 10: scaling dividend and divisor the same way keeps the quotient exactly the same.
3. Compute 14.4 × 10 = 144 and 1.2 × 10 = 12. Now solve 144 ÷ 12 — a whole-number division you already know how to do.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is 3.78 ÷ 0.6?

(A) 0.063
(B) 0.63
(C) 6.3
(D) 63

STEPS TO TRY

1. The divisor 0.6 has one decimal place — shifting both numbers one place makes the divisor a whole number.
2. Multiply both numbers by 10 to build an equivalent problem, then divide, keeping the decimal point in the dividend.
3. 3.78 × 10 = 37.8 and 0.6 × 10 = 6. Now compute 37.8 ÷ 6, keeping the decimal point lined up in your answer.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is 8.4 ÷ 0.7?

(A) 0.12
(B) 1.2
(C) 12
(D) 120

STEPS TO TRY

1. Ask yourself: how many groups of 0.7 fit inside 8.4? Since 0.7 is less than 1, expect MORE groups than 8.
2. Scale both numbers by 10 so you're dividing by a whole number instead of a decimal.
3. 8.4 becomes 84 and 0.7 becomes 7. Solve 84 ÷ 7 — your 7s times-table facts will carry you there.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Why does moving the point the same number of places in both numbers give you the same quotient — and why is it worth doing before you start the divide, multiply, subtract, bring down cycle?

SENTENCE STARTER Multiplying both numbers by ___ doesn't change the quotient because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help